1) Functions and transformations

a) Composite functions

Example 6

The functions f, g, and h are defined by

f: $x \mapsto 4x - 1, x \in \mathbb{R}$ g: $x \mapsto \frac{1}{x+2}, x \neq -2$ h: $x \mapsto (2-x)^2, x \in \mathbb{R}$

Find **a)** fg(x) **b)** hh(x)

a) $\text{fg}(x) = \text{f}\left(\frac{1}{x+2}\right)$ ← Substitute g(x) into fg(x).

$= 4\left(\frac{1}{x+2}\right) - 1$ ← f: × 4 − 1

$= \frac{4 - 1(x+2)}{x+2}$ ← Simplify using a common denominator of $x + 2$.

$= \frac{2-x}{x+2}, x \neq -2$

b) $\text{hh}(x) = \text{h}[(2-x)^2]$

$= [2 - (2-x)^2]^2$ ← h: subtract from 2 and then square.

$= [2 - (4 - 4x + x^2)]^2$

$= (-2 + 4x - x^2)^2$

Example 7

The functions f and g are defined by

f: $x \mapsto ax^2, x \in \mathbb{R}$ g: $x \mapsto 3x + b, x \in \mathbb{R}$ where a and b are constants.

Given that gf(1) = 5 and gg(2) = 14 find

a) the values of a and b **b)** the value of fg(−3).

a) $\text{gf}(1) = \text{g}[a(1)^2] = \text{g}(a)$ ← Substitute $x = 1$ in to f(x).

$= 3a + b$ ← g: multiply by 3 and then add b.

$\text{gf}(1) = 5$

So, $3a + b = 5$

$\text{gg}(2) = \text{g}(6 + b)$ ← Substitute $x = 2$ in to g(x).

$= 3(6 + b) + b$

$\text{gg}(2) = 14$

So, $18 + 4b = 14$ ← Solve for b.

$b = -1$

$3a - 1 = 5$ ← Substitute for $b = -1$ in $3a + b = 5$.

$a = 2$

b) $\text{f}(x) = 2x^2$ $\text{g}(x) = 3x - 1$

$\text{fg}(-3) = \text{f}(3 \times -3 - 1) = \text{f}(-10)$ ← First work out g(−3).

$\text{f}(-10) = 2(-10)^2 = 200$ ← Then substitute this value for x in to f.

b) Inverse functions

Example 9

The functions f is defined by f: $x \mapsto x^2 - 6x + 2$ for $0 \le x \le 3$.

a) Express $f(x)$ in the form $(x + a)^2 + b$, where a and b are constants.

b) State the range of f.

c) State the domain of $f^{-1}(x)$.

d) Find an expression for $f^{-1}(x)$.

e) Express $ff^{-1}(x)$ in terms of x.

f) Sketch on the same diagram the graphs of $y = f(x)$ and $y = f^{-1}(x)$, making clear the relationship between the graphs.

---

a) $x^2 - 6x + 2 = (x - 3)^2 - 9 + 2$

Use the method of completing the square.

$= (x - 3)^2 - 7$

b) Range of f: $-7 \le f(x) \le 2$

When $x = 0$, $f(x) = 2$; when $x = 3$, $f(x) = -7$ (which is the minimum value from **(a)** so f(x) cannot be less than this).

c) Domain of $f^{-1}(x)$: $-7 \le x \le 2$

The domain of f^{-1} is always the same as the range of f.

d) Let $y = (x - 3)^2 - 7$

Let $f(x) = y$.

$x = (y - 3)^2 - 7$

Change x to y and y to x.

Then, $x + 7 = (y - 3)^2$

Hence, $y = 3 \pm \sqrt{x+7}$

Rearrange to get y in terms of x.

$f^{-1}: x \mapsto 3 + \sqrt{x+7}$

However, f^{-1} must be one-to-one, so we must find whether a negative or positive square root is appropriate.

Since $f(0) = 2$, we know that $f^{-1}(2) = 0$.

This means that we need the negative square root giving

$f^{-1}: x \mapsto 3 - \sqrt{x+7}, -7 \le x \le 2$

e) $ff^{-1}(x) = f(3 + \sqrt{x+7})$

$= (3 + \sqrt{x+7} - 3)^2 - 7$

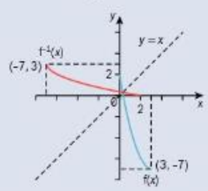
$= (\sqrt{x+7})^2 - 7$

$= x + 7 - 7$

$= x$

$ff^{-1}(x)$ is always the same as x.

f)



$f^{-1}(x)$ is the reflection of $f(x)$ in the line $y = x$.

11. a) The function f: $x \mapsto 2x^2 - 12x + 8$ is defined for $x \in \mathbb{R}$.

  i) Express f(x) in the form $a(x + b)^2 + c$, where a, b and c are constants.

  ii) Find the range of f(x).

b) The function g: $x \mapsto 2x^2 - 12x + 8$ is defined for $x \geq D$.

  i) Find the smallest value of D for which g has an inverse.

  ii) For this value of D, find an expression for $g^{-1}(x)$ in terms of x.

12. The function f is defined by f: $x \mapsto 4x - x^2$ for $x \geq 2$.

a) Express $4x - x^2$ in the form $a - (x - b)^2$, where a and b are positive constants.

b) Express $f^{-1}(x)$ in terms of x.

13. A function f is such that $f(x) = 2 + \sqrt{\frac{x-1}{3}}$, for $x \geq 1$.

a) Find $f^{-1}(x)$ in the form $ax^2 + bx + c$, where a, b and c are constants.

b) Find the domain of f^{-1}.

14. The functions f and g are defined for $x \in \mathbb{R}$ by

f: $x \mapsto 4x - a$ g: $x \mapsto b + ax$

where a and b are constants.

Given that $f^{-1}(1) = 2$ and $g^{-1}(-1) = -3$, find the values of a and b.

c) Transformations

i) Translations

2. a) Sketch the graph of $y = g(x)$, where $g(x) = 2^x$.

b) On separate diagrams, sketch the graphs of

  i) $y = g(x + 2)$ **ii)** $y = g(x) - 1$ **iii)** $y = g(x) + 4$.

Mark on each sketch, where possible, the coordinates of the points where the curve cuts the axes and state the equations of any asymptotes.

3. The graph of $y = f(x)$ is transformed to the graph of $y = f(x + 5)$.

Describe the transformation.

4. $f(x) = x(x + 3)$

a) Find and simplify the equation of the curve $3 + f(x - 1)$.

b) Describe the transformation.

5. The curve $y = x^2 - x + 1$ is translated by $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$.

Find and simplify the equation of the translated curve.

ii) Reflections

For example, to sketch the graph of $-f(9 + x)$, first apply the translation of $f(9 + x)$ and then the reflection. To sketch the graph of $f(4 - x)$, rewrite as $f(-(x - 4))$, reflect in the y-axis then translate 4 to the right.

2. The graph of $y = f(x)$ is transformed to the graph of $y = f(-x) + 7$.

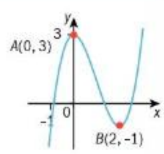
Describe the transformation.

3. The diagram shows a sketch of the curve f(x), which passes through the points $A(0, 3)$ and $B(2, -1)$.

Sketch the graphs of

a) $y = -f(x)$ **b)** $y = f(-x)$ **c)** $y = -f(x + 4)$ **d)** $y = 3 - f(-x)$.

In each case, mark the new position of the points A and B, writing down their coordinates.



4. The points $P(-3, 5)$ and $Q(-2, -8)$ lie on the curve with equation $y = f(x)$.

Find the coordinates of P and Q after the curve has been transformed by the following transformations:

a) $f(-x)$ **b)** $-f(x)$ **c)** $f(-x + 1)$ **d)** $-f(x) - 5$

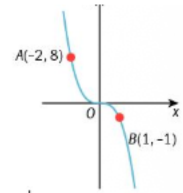
iii) Stretches

2. The diagram shows a sketch of the curve f(x), which passes through the origin, O, and the points $A(-2, 8)$ and $B(1, -1)$.

   On separate diagrams, sketch the graphs of

   a) $y = 4f(x)$ **b)** $y = f(-\frac{1}{2}x)$ **c)** $y = -2f(x + 1)$ **d)** $y = 1 + f(-2x)$.

   Mark the new position of the points O, A and B on each transformation, stating their coordinates.



3. The diagram shows a sketch of the curve f(x).

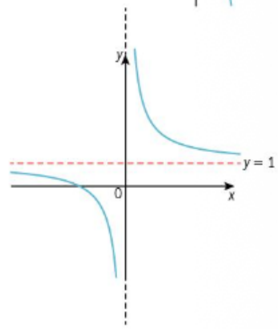
   The curve has a horizontal asymptote with equation $y = 1$ and a vertical asymptote with equation $x = 0$.

   On separate diagrams, sketch the graphs of

   a) $y = f(2x)$ **b)** $y = -3f(x)$ **c)** $y = 2f(x - 3)$

   d) $y = 4 - f(\frac{1}{2}x)$ **e)** $y = 3 + 4f(x)$ **f)** $y = 4f(-\frac{1}{2}x)$.

   Mark on each sketch the equations of the asymptotes and, if possible, any points where the curve cuts the axes.



4. The graph of $y = f(x)$ is transformed to the graph of $y = -f(2x)$.

   Describe fully the two single transformations that have been combined to give the resulting transformation.

Summary:

Chapter summary

Domain and range

- If a function maps x onto f(x), then the values of x is the **domain** of the function and the corresponding values of f(x) is the **range** of the function.

Functions

- A **function** is defined as a mapping where every element of the domain (set x) is mapped onto exactly one element of the range (set y).
- If every element in the domain is mapped onto exactly one element in the range we say the function is a **one-to-one** function.
- If every element in the domain is mapped onto exactly one element in the range, but some elements in the range arise from more than one element in the domain we say the function is a **many-to-one** function.
- We say that each member of a set is an **element** of that set.
- $x \in \mathbb{R}$ means that x is a member of all the real numbers.

Composite functions

- The composite function **fg(x)** means apply g first followed by f.
- The composite function **gf(x)** means apply f first followed by g.

Inverse functions

- The **inverse function** of f, written as $\mathbf{f^{-1}(x)}$, maps the range back onto the domain.
- The steps for finding the inverse function $f^{-1}(x)$ given f(x) are:
  1. Let $f(x) = y$.
  2. Change x to y and y to x.
  3. Rearrange to get y in terms of x.
  4. Write in the correct form.
- The domain of f^{-1} is always the same as the range of f.
- The range of f^{-1} is always the same as the domain of f.
- $ff^{-1}(x)$ and $f^{-1}f(x)$ are always the same as x.
- The graph of $f^{-1}(x)$ is the reflection of the graph of f(x) in the line $y = x$.
- If $f(x) = f^{-1}(x)$, then we say f(x) is a **self-inverse** function. If f is self-inverse, then $ff(x) = x$.

Transformations

- We can transform the graph of a function by moving it horizontally or vertically. This transformation is called a **translation**.
- $\mathbf{f(x + a)}$ is a horizontal translation of $-a$.
- $\mathbf{f(x) + a}$ is a vertical translation of a.

- We can transform the graph of a function by **reflecting** the graph in one of the axes.
- $\mathbf{-f(x)}$ is a reflection in the x-axis.
- $\mathbf{f(-x)}$ is a reflection in the y-axis.
- We can transform the graph of a function by **stretching** the graph horizontally or vertically.
- $\mathbf{af(x)}$ is a stretch with factor $\boldsymbol{a}$ in the y-direction.
- $\mathbf{f(ax)}$ is a stretch with factor $\frac{1}{a}$ in the x-direction.